

11.12

求戴维南阻抗 Z_{Th} , 此时将独立电压源短路, 左侧节点等效为地, 则从 Z_L 两端向外看, 一条支路为 4Ω , 另一条为 $-j3 + (j2 \parallel 5)$

$$\text{由 } j2 \parallel 5 = \frac{j2 \cdot 5}{j2 + 5} = \frac{j10}{5 + j2} = \frac{20 + j50}{29} = 0.6897 + j1.7241,$$

$$\text{则 } -j3 + (j2 \parallel 5) = 0.6897 - j1.2759,$$

$$\text{即 } Z_{Th} = 4 \parallel (0.6897 - j1.2759) = \frac{4(0.6897 - j1.2759)}{4 + 0.6897 - j1.2759} \approx 0.8234 - j0.8642 \Omega$$

$$\text{则 } \text{最大功率传输时, } Z_L = Z_{Th}^* = 0.8234 + j0.8642 \Omega$$

求戴维南电压 V_{Th} , 设负载端节点电压为 V_B , 右侧节点电压为 V_C , 则节点方程,

$$\begin{cases} \frac{V_B - V_A}{4} + \frac{V_B - V_C}{-j3} = 0 \\ \frac{V_C - V_A}{j2} + \frac{V_C}{5} + \frac{V_C - V_B}{-j3} = 0 \end{cases} \quad \text{代入计算得 } V_B = 192.99 - j60.73,$$

$$\text{则 } V_{Th} = V_B, \quad |V_{Th}| = \sqrt{192.99^2 + 60.73^2} = 202.32 \text{ V}$$

$$\text{则 } P_{max} = \frac{|V_{Th}|^2}{4R_{Th}}, \text{ 其中 } R_{Th} = 0.8234 \Omega,$$

$$\text{代入得 } P_{max} \approx 12.43 \text{ kW}$$

11.19

将右侧电阻 R 端口等效为戴维南电路

求 V_{Th} ,

$$Z_1 = 100 + j100, \quad V_{Th} = 100 \angle 0^\circ \cdot \frac{100 + j100}{100 + (100 + j100)} = 100 \cdot \frac{100 + j100}{200 + j100} = 60 + j20$$

$$\text{则 } |V_{Th}| = \sqrt{60^2 + 20^2} = 63.25 \text{ V}$$

求 Z_{Th} :

$$\text{将独立电压源短路: } Z_N = 100 \parallel 100 + j100 = \frac{100(100 + j100)}{100 + (100 + j100)} = 60 + j20$$

$$\text{则 } Z_{Th} = -j100 + Z_N = -j100 + 60 + j20 = 60 - j80 \Omega$$

$$\text{最大功率条件为: } R = |Z_{Th}| = \sqrt{60^2 + 80^2} = \sqrt{3600 + 6400} = 100 \Omega$$

$$\text{则 } R = 100 \Omega$$

$$\text{负载功率为: } P_R = \frac{|V_{Th}|^2 R}{|Z_{Th} + R|^2}, \text{ 代入得 } |V_{Th}|^2 = 63.25^2 = 4000, \quad Z_{Th} + R = (60 - j80) + 100 = 160 - j80$$

$$|Z_{Th} + R|^2 = 160^2 + 80^2 = 25600 + 6400 = 32000$$

$$\text{则 } P_{max} = \frac{4000 \times 100}{32000} = 12.5 \text{ W}$$

11.51

$$(a) Z_T = 2 + (10 - j5) \parallel (8 + j6) = 2 + \frac{(10 - j5)(8 + j6)}{18 + j} = 2 + \frac{110 + j20}{18 + j}$$

$$= 8.152 + j0.768 = 8.188 \angle 5.382^\circ$$

RII pf = $\cos(5.382^\circ) = 0.996$, 由于 Z_T 虚部为正, 整体电路呈感性, 电流滞后电压

$$(b) S = VI^* = \frac{|V|^2}{Z_T^*}, \text{ 代入 } |V| = 50, \text{ 则 } S = \frac{50^2}{8.188 \angle (-5.382^\circ)} = \frac{2500}{8.188} \angle 5.382^\circ = 305.33 \angle 5.382^\circ$$

$$P = |S| \cos \theta = 305.33 \cos 5.382^\circ = 303.99 \text{ W}$$

$$(c) Q = |S| \sin \theta = 305.33 \sin 5.382^\circ = 28.64 \text{ VAR}, \text{ 整体电路呈感性, 所以无功功率为正}$$

$$(d) S = |S| = 305.33 \text{ VA}$$

$$(e) S = P + jQ = 303.99 + j28.64 \text{ VA}$$

11.56

$$-j2 \parallel 6 = \frac{6(-j2)}{6-j2} = \frac{12 \angle -90^\circ}{6.32456 \angle -18.435^\circ} = 1.89736 \angle -71.565^\circ = 0.6 - j1.8$$

$$3 + j4 + [-j2 \parallel 6] = 3.6 + j2.2$$

$$I_o = \frac{3.6 + j2.2}{8.6 + j2.2} (2 \angle 30^\circ) = \frac{4.219 \angle 31.4296^\circ}{8.87694 \angle 14.7493^\circ} (2 \angle 30^\circ) = 0.95055 \angle 47.08^\circ$$

$$V_o = 5I_o = 4.75275 \angle 47.08^\circ, S = V_o I_o^* = (4.75275 \angle 47.08^\circ) (2 \angle -30^\circ)$$

$$= 9.5055 \angle 17.08^\circ$$

$$= 9.086 + j2.792 \text{ VA}$$

11.74

(a)

$$\theta_1 = \cos^{-1}(0.8) = 36.87^\circ, S_1 = \frac{P_1}{\cos \theta_1} = \frac{24}{0.8} = 30 \text{ kVA}, Q_1 = S_1 \sin \theta_1 = 30 \times 0.6 = 18 \text{ kVAR}$$

$$S_1 = 24 + j18 \text{ kVA}$$

$$\theta_2 = \cos^{-1}(0.95) = 18.19^\circ, S_2 = \frac{P_2}{\cos \theta_2} = \frac{40}{0.95} = 42.105 \text{ kVA}, Q_2 = S_2 \sin \theta_2 = 13.144 \text{ kVAR}$$

$$S_2 = 40 + j13.144 \text{ kVA}$$

$$S = S_1 + S_2 = 64 + j31.144 \text{ kVA}, \theta = \tan^{-1}\left(\frac{31.144}{64}\right) = 25.95^\circ, \text{ pf} = \cos \theta = 0.8992$$

(b)

$$\theta_2 = 25.95^\circ, \theta_1 = 0^\circ, Q_c = P [\tan \theta_2 - \tan \theta_1] = 64 [\tan(25.95^\circ) - 0] = 31.144 \text{ kVAR}$$

$$C = \frac{Q_c}{\omega V_{rms}^2} = \frac{31144}{2\pi \times 60 \times 120^2} = 5.74 \text{ mF}$$